

CMAN :=

Given data $N = 32$, FP32 = 4 bytes , DRAM bandwidth = 320 GB/s

Compute = 10 TFLOPS

Task 1:

$N = 32$ times , So each $B[K][J]$ accessed 32 times

- A accesses : $N^3 = 32^3 = 32,768$ elements
- B accesses : $N^3 = 32^3 = 32,768$ elements

Total elements accessed :

$$2N^3 = 2 \times 32^3 = 65,536$$

Total DRAM bytes :

$$65,536 \times 4 = 262,144 \text{ bytes} = 256 \text{ kB}$$

Total DRAM traffic (naive) = 256 kB

Task 2:

Tiled size $T = 8$

$$(N/T)^2 = (32/8)^2 = 16 \text{ tiles per matrix}$$

$$\begin{aligned} \text{Each tile holds } T^2 &= (8)^2 = 64 \text{ elements} \\ &= 64 \times 4 \text{ bytes} = 256 \text{ bytes} \end{aligned}$$

$$\text{DRAM loads for A} = N^2 \times 4$$

$$= (32)^2 \times 4 = 4,096 \text{ bytes}$$

Total DRAM for A and B

$$= N^2 \times 4 \times 2$$

$$= 4,096 \times 2 = 8,192 \text{ bytes}$$

DRAM traffic = total DRAM load for A and B

DRAM traffic (tiled) = 8,192 bytes

Task 3:

Naive traffic = 262,144 bytes

Tiled traffic = 32,768 bytes

Traffic ratio :

$$\text{Naive / Tiled} = 262,144 / 32,768 = 8 \text{ (T)}$$

If we consider only B's reuse :

$$\text{Tiled traffic} = 262,144 / 32 = 8,192 \text{ bytes}$$

Ratio :

$$262,144 / 8,192 = 32 = N$$

Explanation:

In the naive matrix multiplication, matrix B is entirely re-read from DRAM for each row of matrix A, since every output element requires a full traversal of B. Consequently, matrix B is reloaded N times across the entire computation. In the tiled approach, each $T \times T$ sub-block of B is loaded from DRAM exactly once and retained in on-chip shared memory for reuse within that tile. Therefore, naive DRAM traffic is exactly N times larger than tiled DRAM traffic.

Task 4:

Bandwidth = 320 GB/s

Compute = 10 TFLOPS

$$\text{Total FLOPs} = 2 \times N^3 = 2 \times 32^3 = 65,536 \text{ FLOPs}$$

Naive Case:

$$\text{Memory time: } t_{\text{memory}} = \text{Traffic} / \text{BW}$$

$$= 262,144 / (320 \times 10^9)$$

$$= 8.192 \times 10^{-7} \text{ s} = 0.819 \text{ ms}$$

$$\text{Compute time: } t_{\text{compute}} = \text{FLOPs} / P$$

$$= 65,536 / 10^{13}$$

$$= 6.554 \times 10^{-9} \text{ s} = 0.00655 \text{ ms}$$

$$t_{\text{memory}} (0.819 \text{ ms}) \blacksquare t_{\text{compute}} (0.00655 \text{ ms})$$

Execution time (naive) : 0.819 ms

Bottleneck: Memory-bound.

Tiled Case (T=8):

Memory time:

$$t_{\text{memory}} = 32,768 / (320 \times 10^9)$$

$$= 1.024 \times 10^{-7} \text{ s} = 0.1024 \text{ ms}$$

Compute time: $\text{FLOPs} / P = 65,536 / (1.0 \times 10^{13})$

$$\Rightarrow 6.554 \times 10^{-9} \text{ s} = 0.00655 \text{ ms}$$

$t_{\text{memory}} (0.1024 \text{ ms}) > t_{\text{compute}} (0.00655 \text{ ms})$

Execution time (tilted) = 0.1024 ms

Bottleneck: Memory-bound.